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Convexity Tests

- f convex $\iff \text{epi}(f)$ convex
- Twice diff: $\nabla^2 f \succeq 0 \ \forall x$
- 1st-order: $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$
- Monotone grad: $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0$
- f convex $\implies e^f$ convex; pointwise max of convex = convex
- Not** convex: unique min, single local min, product of convex, $\sqrt{x}$ on $[0, 1]$ (Lip const. nonexistent)
- Quadratic $x^\top Qx$: convex $\iff Q \succeq 0$ (check eigenvalues)
- Separating hyperplane: disjoint convex $C_1, C_2 \implies \exists a \neq 0, b$ s.t. $a^\top x + b \leq a^\top y + b \ \forall x \in C_1, y \in C_2$

Conjugate f^*

$$f^*(y) = \sup_x (y^\top x - f(x))$$

- $f = f^{**}$ if proper, convex, closed epi
- Ex: $f(x) = -x^2$ on $[-1, 1] \rightarrow f^* = 1 + |y|$
- Ex: $f(x) = -\ln x \ (x > 0) \rightarrow f^*(y) = -1 - \ln(-y) \ (y < 0)$
- Ex: $f(x) = e^x \rightarrow f^* = y \ln y - y \ (y > 0)$

KKT & Duality Lagrangian $\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^\top g(x) + \nu^\top h(x)$

Dual: $\max_{\lambda \geq 0, \nu} d(\lambda, \nu), d = \inf_x \mathcal{L}$

Weak duality always: $d \leq p^*$

Strong duality + KKT (convex + Slater $\exists \tilde{x} : g(\tilde{x}) < 0$):

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) &= 0 \\ g(x^*) &\leq 0, \ h(x^*) = 0, \ \lambda^* \geq 0 \\ \lambda^{*\top} g(x^*) &= 0 \end{aligned}$$

(Sufficiency: any KKT pt. is optimal)

Subdifferential $0 \in \partial f(x^*) \iff x^* \min(|x| : \partial f(0) = [-1, 1])$

Problem Classes $\text{LP} \subset \text{QP} \ (P \succeq 0) \subset \text{SOCP} \subset \text{SDP}$

1st-order Methods & Rates ($\kappa = L/\mu$)

- GD (smooth μ -sc): $N \sim \kappa \ln(1/\varepsilon)$
- Nesterov: $N \sim \sqrt{\kappa} \ln(1/\varepsilon)$

- Smooth convex: Nesterov $N \sim \sqrt{LR^2/\varepsilon}$
- Lipschitz convex (subgrad): $N \sim (RL_f/\varepsilon)^2$
- Proj. GD = GD rate if prox cheap (ℓ_1 , ball, simplex)

Tactics

- Convexity MCQ/TF: test Hessian/1st-order; give counter-ex. (e.g. $f(x) = |x|$ on $[-1, 1]$ not convex)
- Conjugate: use def., split cases on argmax (domain, stationary pt.)
- Dual/KKT: write $\mathcal{L}$, set $\nabla_x = 0$, solve for λ^* via comp. slackness
- Rates: identify μ, L (or κ); remember $1/N^2, 1/\sqrt{N}$

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Convexity (Prob 1 every year)

Def. $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ convex $\iff \text{epi}(f) = \{(x, t) : f(x) \leq t\}$ convex set $\iff$

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \ \forall \theta \in [0, 1].$$

Characterizations (use these!)

- Twice diff.: $\nabla^2 f(x) \succeq 0 \ \forall x$ (Hessian PSD; check eigenvalues for quadratics $x^\top Qx$).
- 1st-order: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) \ \forall x, y$.
- Monotone gradient: $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0$.
- Restriction to any line $t \mapsto f(x + tv)$ convex.
- $f^{**} = f$ (proper, closed epi).

Sets: Intersection, affine image/preimage, half-spaces, polyhedra, balls, $\mathbb{S}_+^n$ convex. Separating hyperplane thm always holds for disjoint closed convex C_1, C_2 .

Operations preserving conv.: nonnegative lin. comb., pointwise max/sup, composition w/ affine, indicator ψ_C .

Does NOT imply convex (common MC/TF traps):

- Unique/global minimizer (counterex: $f(x) = |x|$ on $[-1, 1]$, 1 outside – local min at 0 but not convex).
- Single local min = global min.
- $\nabla^2 f$ bounded below.
- $|\nabla f(x) - \nabla f(y)| \geq \mu|x - y|$ (strong mono $\implies$ strong conv).
- Pointwise min of convex f_i (counterex: $f_1 = (x + 1)^2, f_2 = (x - 1)^2$).
- Product $f_1 f_2$ (counterex: $x \cdot (-x) = -x^2$ concave).
- $g(x) = \exp(f(x))$ if f convex (not always).
- $f(x) = 1/\ln x \ (x > 1)$ – check Hessian.

Examples: $1/x$ convex ($x > 0$), $\sqrt{x}$ on $[0, 1]$: $\nabla f \rightarrow \infty$ at 0 $\implies$ **no finite Lip const.** $\frac{1}{2}x^2 + p \ln x$ convex on $[1, \infty)$ iff $p \leq 1$ (Hessian $1 - p/x^2 \geq 0$).

Tactic (MC/TF): Test Hessian or give counterex. Check proper/closed epi for $f = f^{**}$.

Conjugates $f^*(y) = \sup_x (y^\top x - f(x))$

Prop.: f proper convex closed epi $\implies f^{**} = f$. $y \in \partial f(x) \iff x \in \partial f^*(y)$.

Compute: Solve unconstrained max of $yx - f(x)$ (set deriv=0), check if in domain; else evaluate at boundaries. Cases often needed.

Quiz examples:

- $f(x) = -x^2, x \in [-1, 1]$: $f^*(y) = 1 + |y|$ (V-shape plot).
- $f(x) = -\ln x \ (x > 0)$: $f^*(y) = -1 - \ln(-y)$ for $y < 0, +\infty$ else.
- $f(x) = e^x$: $f^*(y) = y \ln y - y \ (y > 0), +\infty$ else.
- $f(x) = x^2 + x, x \in [-1, 1]$:

$$f^*(y) = \begin{cases} -y & y \leq -1, \\ \frac{1}{4}(y - 1)^2 & -1 \leq y \leq 3, \\ y - 2 & y \geq 3. \end{cases}$$

Tactic: Stationary point $x = y/2$ or similar; piecewise linear/quadratic result common. Plot for 2023-style.

Subdifferentials & Mns

$$\partial f(\bar{x}) = \{v : f(x) \geq f(\bar{x}) + v^\top (x - \bar{x}) \ \forall x\}.$$

Optimality: x^* minimizer $\iff 0 \in \partial f(x^*)$.

Example: $f(x) = |x|$,

$$\partial f(x) = \begin{cases} \{-1\} & x < 0, \\ [-1, 1] & x = 0, \\ \{1\} & x > 0. \end{cases}$$

Minimizer where $0 \in \partial f$ (at B in plotted subdiff questions).

Duality, Lagrangian, KKT

Lagrangian: $\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^\top g(x) + \nu^\top h(x), \lambda \geq 0$.

Dual fn: $d(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu) \leq f^*$ (weak duality *always*).

Strong duality: Convex + Slater ($\exists \tilde{x}$ strict ineq. feasible) $\implies d^* = f^*$.

KKT (nec. & suff. under convex+Slater):

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) &= 0 && \text{(stat.)} \\ g(x^*) &\leq 0, \ h(x^*) = 0 && \text{(primal feas.)} \\ \lambda^* &\geq 0 && \text{(dual feas.)} \\ \lambda^{*\top} g(x^*) &= 0 && \text{(compl. slack.)} \end{aligned}$$

2023 SVM dual ex.: $\max_{\lambda \geq 0} 2\lambda_B - 4\lambda_B^2 \implies \lambda_A = \lambda_B = 1/4, a^* = (1/2, 1/2)^\top, b^* = -1$. Hyperplane plot through midpoint.

2024 ex.: $f = 1/x_1 + 4/x_2$, box constraints; solve KKT explicitly for x^*, p^* .

Tactic: Write full Lagrangian $\rightarrow$ dual $\rightarrow$ solve small QP. Check Slater for suff.

Convex Problem Classes

Hierarchy: $\text{LP} \subset \text{QP} \subset \text{SOCP} \subset \text{SDP}$.

- LP:** lin. obj + lin. ineq/eq.
- QP:** $\frac{1}{2}x^\top Px + q^\top x \ (P \succeq 0)$, lin. constr.
- QCQP:** quad. obj + quad. constr.
- SOCP:** lin. obj + $\|A_i x + b_i\| \leq c_i^\top x + d_i$.
- SDP:** $\langle C, X \rangle, X \succeq 0$, lin. matrix ineq.

Common trap (2024): $|A_i x + b_i|^2 \leq c^\top x + d_i$ is SOCP (not always QP/SDP).

Algorithms & Rates (Large-Scale)

μ -strong conv., L -smooth $\implies \kappa = L/\mu$.

Convergence $(f(x_N) - f(x^*)) \leq \varepsilon$:

Projected GD: same rates (prox nonexpansive). Prox onto ℓ_∞ -ball, simplex, ℓ_1 -ball cheap.

Lipschitz const: smallest L s.t. $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ (or for f). $\sqrt{x}$ on $[0, 1]$: **does not exist**.

Tactic: Identify μ, L from Hessian eigenvalues. For $\mu = 0$ regularize or use accel. rate $O(1/N^2)$. Plot directions/hyperplanes when asked.

General tips:

- Convexity: Hessian/counterex.
- Conjugate/KKT: explicit deriv + cases.
- Rates: match assumptions to table.
- Always weak duality. $f = f^{**}$ needs proper+closed epi.

Common funcs: $\ln, e^x, 1/x, \text{quadratics, norms}$.

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1. Convex Sets & Functions

Convex Set: $x, y \in C, \theta \in [0, 1] \implies \theta x + (1 - \theta)y \in C$. **Preserving Set Convexity:** Intersection (any), Affine map $f(x) = Ax + b$ (image and pre-image), Minkowski sum $C_1 + C_2$. **Convex Function:** $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$. $\iff$ **Epigraph** $\{(x, t) \mid f(x) \leq t\}$ is a convex set. **Properties:**

- f convex $\implies$ Local min is Global min.
- Jensen:** $f(E[x]) \leq E[f(x)]$.
- 1st Order:** $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$.
- 2nd Order:** $\nabla^2 f(x) \succeq 0$ (PSD).
- Strict Convexity:** $\nabla^2 f(x) \succ 0 \implies$ unique minimizer.

Preserving Function Convexity:

- $w_i f_i$ with $w_i \geq 0$.
- Pointwise **MAX:** $g(x) = \max(f_1(x), f_2(x))$.
- Affine composition: $f(Ax + b)$.
- Note:* Pointwise MIN is usually NOT convex.

2. Conjugate Function $f^*(y)$

$f^*(y) = \sup_{x \in \text{dom}f} \{y^\top x - f(x)\}$. **Properties:** f^* is always convex (even if f isn't). If f is closed, proper, convex: $f^{**} = f$. **Common Pairs:**

- $f(x) = ax + b \implies f^*(a) = -b$, else ∞ .
- $f(x) = \frac{1}{2}x^\top Qx \implies f^*(y) = \frac{1}{2}y^\top Q^{-1}y$.
- $f(x) = |x| \implies f^*(y) = 0$ if $|y| \leq 1$, else ∞ .
- $f(x) = -\ln x \implies f^*(y) = -1 - \ln(-y)$ for $y < 0$.
- $f(x) = e^x \implies f^*(y) = y \ln y - y$ for $y > 0$.
- $f(x) = I_C(x)$ (ind.) $\implies f^*(y) = \sigma_C(y)$ (supp.).

Dual trick: $y \in \partial f(x) \iff x \in \partial f^*(y) \iff f(x) + f^*(y) = y^\top x$.

3. Subdifferentials

$\partial f(x) = \{g \mid f(y) \geq f(x) + g^\top (y - x) \ \forall y\}$. **Optimality:** x^* is global min $\iff 0 \in \partial f(x^*)$. **Examples:**

- $f(x) = |x| \implies \partial f(0) = [-1, 1]$.
- $f(x) = \max(f_1, f_2) \implies \partial f(x) = \text{conv}(\cup \partial f_i(x))$.

4. Duality & KKT

Primal: $\min f(x)$ s.t. $g_i(x) \leq 0, h_j(x) = 0$. **Lagrangian:** $\mathcal{L}(x, \lambda, \nu) = f(x) + \sum \lambda_i g_i(x) + \sum \nu_j h_j(x)$. **Dual Function:** $d(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$. **Weak Duality:** $d^* \leq f^*$ always. **Strong Duality:** $d^* = f^*$. Holds if: 1. Convex problem + **Slater's Condition** ($\exists x : g_i(x) < 0$ for all non-affine g_i). **KKT Conditions (Stationarity, Primal Feas, Dual Feas, Comp. Slackness):** 1. $\nabla f(x^*) + \sum \lambda_i^* \nabla g_i(x^*) + \sum \nu_j^* \nabla h_j(x^*) = 0$. 2. $g_i(x^*) \leq 0, h_j(x^*) = 0$. 3. $\lambda_i^* \geq 0$. 4. $\lambda_i^* g_i(x^*) = 0$.

5. Problem Classes

LP: Linear obj, linear constraints. **QP:** Quad obj $P \succeq 0$, linear constraints. **QCQP:** Quad obj, quad constraints $P_i \succeq 0$. **SOCP:** Linear obj, cone: $\|Ax + b\|_2 \leq c^\top x + d$. **SDP:** Linear obj, LMI: $x_1 F_1 + \dots + x_n F_n + G \preceq 0$. **Hierarchy:** $\text{LP} \subset \text{QP} \subset \text{QCQP} \subset \text{SOCP} \subset \text{SDP}$.

6. Smoothness & Strong Conv.

L -Smooth: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$. $\iff f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|x - y\|^2$. $\iff \nabla^2 f(x) \preceq LI$. **μ -Strongly Convex:** $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|x - y\|^2$. $\iff \nabla^2 f(x) \succeq \mu I$. **Condition Number:** $\kappa = L/\mu \geq 1$.

7. Algorithms

Gradient Descent (GD): $x_{k+1} = x_k - T \nabla f(x_k)$.

- L -smooth, μ -str. conv: $T = \frac{2}{L+\mu}$. Dist to opt: $\|x_k - x^*\| \leq (\frac{\kappa-1}{\kappa+1})^k \|x_0 - x^*\|$. Rate: **Linear** $O(\kappa \ln(1/\epsilon))$.
- L -smooth only: $T = 1/L$. Rate: $O(1/k)$.

Nesterov Accelerated:

- L -smooth, μ -str. conv: Rate: $O(\sqrt{\kappa} \ln(1/\epsilon))$.
- L -smooth only: Rate: $O(1/k^2)$.

Subgradient Method: $x_{k+1} = x_k - T_k g_k, g_k \in \partial f(x_k)$.

Rate: $O(1/\epsilon^2)$ iterations ($O(1/\sqrt{k})$ convergence). **Projected**

GD (PGD): $x_{k+1} = \text{proj}_C(x_k - T \nabla f(x_k))$. $\text{proj}_C(z) = \operatorname{argmin}_{x \in C} \|x - z\|^2$.

- ℓ_2 ball: $z / \max(1, \|z\|_2)$.
- ℓ_∞ ball: $\text{clamp}(z, -1, 1)$ element-wise.
- ℓ_1 ball: Soft-thresholding.

8. Useful Formulas

Matrix Calculus: $\nabla_x(a^T x) = a, \nabla_x(x^T A x) = (A + A^T)x$.

$\nabla_x^2(x^T A x) = A + A^T$. **Quadratics:** $f(x) = \frac{1}{2} x^T Q x + b^T x + c$.

$\nabla f = Qx + b, \nabla^2 f = Q$. Min at $x^* = -Q^{-1}b$. **SVM**

Hyperplane: $\min \frac{1}{2} \|a\|^2$ s.t. $y_i(a^T x_i + b) \geq 1$. Dual:

$\max \sum \lambda_i - \frac{1}{2} \sum \lambda_i \lambda_j y_i y_j x_i^T x_j$.